

Nazmus Sakib Abir

AI/ML Engineer | Technical Lead

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PROFESSIONAL SUMMARY

Applied AI/ML engineer with professional experience since 2018, building practical systems across computer vision, generative AI, and multimodal RAG. Combines technical leadership with hands-on delivery across model research, Python services, product interfaces, cloud deployment, and operational optimization.

CORE STRENGTHS

End-to-end AI delivery: Model research and evaluation, computer vision, generative AI, multimodal RAG, REST APIs, product interfaces, deployment, and operations.

Technical leadership: Project proposal, team guidance, demos, research documentation, mentoring, and stakeholder communication under real hardware, data, licensing, and time constraints.

PROFESSIONAL EXPERIENCE

Chowa Giken Corporation | Japan | Oct 2022 - Present

AI/ML Engineering | Technical-Development Leadership

- Own applied research and product development across multimodal RAG, generative AI, computer vision, backend services, interfaces, and cloud deployment.
- Propose and lead technical-development initiatives from model investigation and **proof of concept through demonstrations and production-oriented delivery**.
- Develop and optimize AI pipelines under GPU memory, latency, data availability, commercial-use, and delivery-time constraints.
- Lead or contribute directly to document intelligence, explainable forecasting, generative image workflows, anomaly detection, and AI demo platforms.
- Contributed to **three patent applications**, with **lead contributions to two**; also support technical documentation, mentoring, and external presentations.

MononAI Limited | Jun 2018 - 2022

Full Stack Developer | Deep Learning Engineer

- **Built the entire production AI and backend** for a real-time multi-camera face-intelligence product.
- Built computer-vision systems spanning face recognition, vehicle detection, ANPR, REST APIs, databases, caching, and WebSocket interactions.
- Improved **recognition accuracy and multi-camera throughput**, query performance, and application responsiveness while expanding into full-stack product delivery.
- Delivered frontend features and workflow improvements, including a substantial registration-system redesign.

PROJECT HIGHLIGHTS

Visual Response RAG | 2024 - 2026

Proposed and led **two generations of document-chat systems** that combine written answers with relevant source visuals; **contributed to a related patent application** alongside hands-on engineering, demos, and team guidance.

RAG | OCR | LLMs | Technical leadership | Patent application

Forecast Explainability | 2024

Led development of an AI system that provides useful context around forecast-model predictions, spanning solution design, demonstration, and **a contribution to a related patent application**.

Explainable AI | LLMs | Technical leadership | Patent application

PROJECT HIGHLIGHTS (CONTINUED)

Vision Inference Optimization | 2023

Optimized a 360-degree image-processing pipeline in about one week, making it **more than three times faster** without retraining or changing its outputs.

[Computer vision](#) | [PyTorch](#) | [Performance](#)

Generative Floorplan Development | 2025 - 2026

Led first-stage base-floorplan preparation for constrained floorplan generation, then drove model development and evaluation, REST API delivery, deployment, and **contributed to a related patent application**.

[Generative AI](#) | [Image generation](#) | [FastAPI](#) | [Patent application](#)

Product Anomaly Detection | 2023 - 2024

Led Bangladesh engineering contributions across backend and React interface development for data upload, model training, inference, and product performance improvements.

[Anomaly detection](#) | [React](#) | [Backend](#)

Multi-Camera Face Intelligence | 2018 - 2022

Built the entire production AI and backend system for real-time CCTV face monitoring, then expanded into performance optimization and full-stack product improvements.

[Face recognition](#) | [Real-time systems](#) | [Backend ownership](#)

Generative House Exterior | 2025 - 2026

Developed model workflows for generating **consistent house-exterior views from elevation plans** while following supplied textures and design requirements.

[Image generation](#) | [Multi-view](#)

TECHNICAL SKILLS

AI and ML: Computer Vision, Face Recognition, Anomaly Detection, Object Detection, Object Tracking, Segmentation, Video Analytics, ANPR, Generative AI, Multimodal RAG, OCR.

Frameworks: Python, PyTorch, OpenCV, Hugging Face Diffusers, PEFT, TensorRT, vLLM, LangChain, LangGraph.

Engineering: FastAPI, Flask, REST APIs, WebSockets, React, Next.js, JavaScript, Docker, Linux, Git, Shell.

Cloud and data: Google Cloud, Azure, Nginx, PostgreSQL, MongoDB, SQLite, Redis.

EDUCATION AND CERTIFICATION

BSc, Electrical and Electronic Engineering: BRAC University | 2014 - 2018 | CGPA 3.67

Google Cloud Associate Cloud Engineer: Completed by 2025

PUBLIC WORK AND LEADERSHIP

- Authored the 2024 technical article, "Combating the Malicious Use of AI-Powered Image Editing."
- Contributed to public project overviews and a demonstration for Visual Response RAG and Forecast Visualization AI.
- Since 2025, helped conduct **two annual university AI teaching and mentoring cohorts** at BRAC University.
- Prepared **approximately one-third of the first cohort's lecture material** and delivered sessions shared with BRAC University and North South University students.
- Supported a mentored group that **won a program final prize**, alongside research documentation and junior-engineer support.